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IS722T2I

Seventh Semester B.E. Degree Examination, December 2025

**SOFTWARE TESTING
(MODEL QUESTION PAPER)**

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Q. No.		Questions	Marks	BL/CO
Module I				
1	a.	Describe the triangle problem statement along with flowchart for traditional implementation.	8	CL2/CO1
	b.	Describe the fault based adequacy criteria in detail. Analyze the different terminologies of mutation based testing.	7	CL2/CO1
	c.	Explain various Test Metrics that exists in software testing.	5	CL2/CO1
OR				
2	a.	Describe the operation of SATM system with detailed screens.	8	CL2/CO1
	b.	Explain Program behavior insights from a Venn diagram for functional testing and Structural testing.	7	CL2/CO1
	c.	Explain Error and fault taxonomies.	5	CL2/CO1
Module II				
3	a.	Demonstrate how the Normal and worst case robust boundary value analysis testing can be done with input domain diagrams for a function of two variables.	10	CL3/CO2
	b.	Write a program to solve the triangle problem.Derive test cases for program based on decision table approach.	10	CL3/CO2
OR				
4	a.	Write a program to solve commission problem. Illustrate it from the perspective of equivalence class testing and derive the test cases.	10	CL3/CO2
	b.	List the assumptions made in fault based testing and illustrate the mutation analysis with sample program.	10	CL3/CO2
Module III				
5	a.	Illustrate the program in C to solve the commission problem and show it from the perspective of dataflow testing. Derive du-Paths for locks, total locks, sales and commission.	10	CL3/CO3
	b.	Illustrate the test oracles and self-check as oracles with a neat diagram.	10	CL3/CO3
OR				
6	a.	Using McCabe’s strongly connected graph, write the path/edge traversal with Cyclomatic Complexity.	10	CL3/CO3



	b.	Apply the DD- graph for the triangle program and condense it with respect to the structured programming constructs.	10	CL3/CO3
Module IV				
7	a.	Demonstrate the importance of Software Testing. Also Analyze how Test Automation is related to Many other Aspects of Software Development.	10	CL3/CO4
	b.	Apply the First Attempt from Manual to Automated Testing with respect to the concept of Record and Playback.	10	CL3/CO4
OR				
8	a.	Illustrate the First Attempt from Manual to Automated Testing with respect to the concept of Record and Playback.	10	CL3/CO4
	b.	Demonstrate the importance of IDE and Programming language in the AutomatedTesting.	10	CL3/CO4
Module V				
9	a.	Illustrate how Selenium web driver works and also show how it interacts with a browser and executes the command.	10	CL3/CO5
	b.	Illustrate the history and evolution of Selenium also highlight the key milestones and their impact on the testing community.	10	CL3/CO5
OR				
10	a.	Demonstrate how to locate elements in the web page using CSS selectors and XPath.	10	CL3/CO5
	b.	Illustrate the different strategies for Locating Elements on the web page using Selenium web driver with an examples.	10	CL3/CO5

Cognitive Levels of Bloom's Taxonomy

No.	CL1	CL2	CL3	CL4	CL5	CL6
Level	Remember	Understand	Apply	Analyze	Evaluate	Create

Course Outcomes

CO1	Understand the concepts of testing and apply to derive different test cases	CL2
CO2	Demonstrate and compare the different testing techniques	CL3
CO3	Apply the appropriate testing techniques in classifying the problem	CL3
CO4	Apply the different approach in software test automation problems and solutions.	CL3
CO5	Apply the techniques and skills on how to use modern software testing tools to support software testing projects.	CL3